

Note the quiz is front & back**SHOW YOUR WORK**Points: _____/16_{16.5}

1. Bonus. Describe what ancient civilizations used caltrops for, which have a tetrahedral geometry. (0.5 pts)

2. ~ A4.104. Provide the correct chemical formula for trinitrogen pentaoxide. (1 pt)

3. A5.2. How many valence electrons does carbon (C) have? (1 pt)

4. ~ ICE. Which of the following is the correct Lewis symbol for the bromide ion? Circle it & write in its charge. (1 pt)



a. Draw/write the Lewis symbol for **nitrogen** using dots to show the required electrons. (1 pt)

5. Draw the Lewis structure of HCN. (1.5 pts)

a. What is the total number of valence electrons in the above formula? (0.5 pts)

6. A5.114. Draw the Lewis structure for an amide anion (NH_2^-). (1 pt)

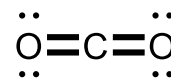
a. Identify the formal charge for all atoms that have non-zero formal charges. (0.5 pts)

7. Draw the Lewis structure of PO_4^{3-} and then calculate the appropriate formal charge for all atom(s) that have a non-zero formal charge. (1.5 pts)

a. Do all atoms in the above problem follow or are below the octet rule? **Yes** or **no**. Circle the correct bolded term. You must have a Lewis structure to be able to answer this question. (0.5 pts)

8. A5.125. Draw the Lewis structure of water (H_2O) and then determine its electron-pair (electron domain) and molecular geometries. Note the geometries *may* be the same. (2 pts)

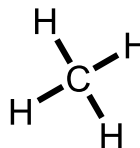
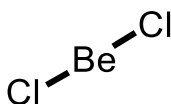
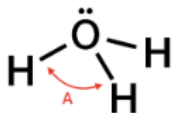
9. ~A5.158. The Lewis structure for CO_2 is provided. Does this molecule possess polar covalent bond(s)? (0.5 pts)



a. Is this molecule polar or nonpolar? Briefly explain your answer. (1 pt)

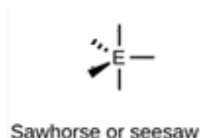
b. Provide the correct IUPAC name for CO_2 . (0.5 pts)

10. For the species below, rank the bond angles from smallest to largest (3). Note they may be drawn in a fashion that doesn't correspond to VSEPR theory. Briefly explain a portion of your answer (1.5 pts)



11. Identify how many atoms and lone pairs are present for a central atom (E) with the electron pair geometry provided below. (1 pt)

Atoms



Lone Pairs

Period

Group

1

18

1 H 1.008 hydrogen																	2 He 4.003 helium																		
																		13	14	15	16	17													
																		5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine													
																		13 Al 26.98 aluminum	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.06 sulfur	17 Cl 35.45 chlorine													
																		11 Na 22.99 sodium	12 Mg 24.31 magnesium											18 Ar 39.95 argon					
																		3 Li 6.94 lithium	4 Be 9.012 beryllium											20 Ca 40.08 calcium					
																		19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.63 germanium	33 As 74.92 arsenic	34 Se 78.97 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton
																		37 Rb 85.47 rubidium	38 Sr 87.62 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.95 molybdenum	43 Tc [97] technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon
																		55 Cs 132.9 cesium	56 Ba 137.3 barium	57-71 La-Lu *	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.8 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po [209] polonium	85 At [210] astatine	86 Rn [222] radon
																		87 Fr [223] francium	88 Ra [226] radium	89-103 Ac-Lr **	104 Rf [267] rutherfordium	105 Db [270] dubnium	106 Sg [271] seaborgium	107 Bh [270] bohrium	108 Hs [277] hassium	109 Mt [276] meitnerium	110 Ds [281] darmstadtium	111 Rg [282] roentgenium	112 Cn [285] copernicium	113 Nh [285] nihonium	114 Fl [289] flerovium	115 Mc [288] moscovium	116 Lv [293] livermorium	117 Ts [294] tennessine	118 Og [294] oganesson

Periodic Table of the Elements

Atomic number → 1

Symbol → **H**

Atomic mass → 1.008

Name → hydrogen

Color Code

- Metal
- Metalloid
- Nonmetal
- Solid
- Liquid
- Gas